

Real Analysis Requests

What the majority of real bacteria/virus questions actually look like

54.6% of 5,250 real Biostars questions (bacteria/virus/pathogen-related) are "analysis request" shaped — data described, a goal stated. Here are ten of them, in full, unedited.

Source: Luna, A. (2023). *BioStars Posts API Output*. Zenodo.

<https://doi.org/10.5281/zenodo.7813785> — CC BY 4.0

Example 1 — Bacterial SNP Detection (1/2)

Biostars #96189 · March 26, 2014 · biostars.org/p/96189

Hi all,

I am working on 10 bacterial genomes (1 reference and 9 mutant) sequenced by Illumina technology. My main aim is to find SNPs that are common in 9 genomes but absent in reference genomes. In last, I would like to do the automatic annotation of those SNPs. Until now, I have done the following steps and wondering if I am on the right path.

First: Extracted the common SNPs in 9 mutant genomes

```
vcf-isec -n +9 -f 1.vcf.gz 2.vcf.gz 3.vcf.gz 4.vcf.gz 5.vcf.gz 6.vcf.gz 7.vcf.gz 8.vcf.gz 9.vcf.gz | bgzip -c > isec1.vcf.gz
```

Second: tab index

```
tabix -p vcf isec1.vcf.gz
```

(continued on next slide — full post, unedited)

Example 1 — Bacterial SNP Detection (2/2)

Biostars #96189, continued

Third: Extracted SNPs present in isec1.vcf.gz but absent in reference strain

```
vcf-isec -c -f isec1.vcf.gz reference.vcf.gz > isec2.vcf
```

Four: Automatic annotation of isec2.vcf. Used snpEFF:

```
java -jar snpEff.jar eff -no-downstream -no-upstream -no-utr -no-intergenic -v database isec2.vcf
```

Most of the SNPs were observed in intergenic region. Should I include these intergenic SNPs or not? Any other suggestions of selecting SNPs.

Regards, Nitin

Example 2 — Viral Nanopore Read Downsampling

Biostars #398159 · September 11, 2019 · biostars.org/p/398159

Tools for downsampling/resampling/subsampling Oxford Nanopore reads

I have got ~40000 MinION reads derived from a virus, whose read length ranges from ~100 to ~34.9k. I would like to re-sample these reads to evaluate the minimum read number enough to detect this virus. As Oxford nanopore reads have no identical read length, I have to take the varying read length into consideration when re-sampling the reads. Are there open source tools available for this task?

Example 3 — Comparing Bacterial Plasmids

Biostars #457131 · August 24, 2020 · biostars.org/p/457131

Hi all,

I got the draft genome of two isolates from WGS sequencing, and then split the contigs belonging to chromosome and plasmid separately. Now, I'd like to compare the plasmids of two isolates, get the information such as similarity, how can I do with that? Any suggestions would be appreciated greatly.

Example 4 — Bacterial Genome Assembly (Nanopore)

Biostars #469465 · October 26, 2020 · biostars.org/p/469465

Hi,

I am doing genome assembly of different bacterial strains with barcoded nanopore sequenced reads. The genome size of the bacteria is 3.8Mb. After genome assembly, I am getting 2 contigs with longest contig length of 3.6 Mb for one strain and 9 contigs with 1.5 Mb contig length for another strain. Actually, the total number of reads for the strain for which I got shortest contig length was almost double that the other strain.

1. Does another round of sequencing with nanopore improve the sequence assembly quality?
2. What could be the reason of getting shortest contig length for the genome for which I had large number of reads?

Thanks

Example 5 — SARS-CoV-2 Spike Protein Sequences

Biostars #9500470 · December 6, 2021 · biostars.org/p/9500470

I want to download Spike protein sequences of 11 variants of the corona:

Alpha (B.1.1.7 and Q lineages) · Beta (B.1.351 and descendent lineages)
· Gamma (P.1 and descendent lineages) · Epsilon (B.1.427 and B.1.429)
· Eta (B.1.525) · Iota (B.1.526) · Kappa (B.1.617.1) · Mu (B.1.621,
B.1.621.1) · Zeta (P.2) · Delta (B.1.617.2 and AY lineages)
· Omicron (B.1.1.529)

From where I could find and download these variant spike protein sequences?

Example 6 — Draft Viral Genome from De Novo Contigs

Biostars #418459 · January 25, 2020 · biostars.org/p/418459

Hi, I am working on the viral genome (single negative-stranded RNA virus) and the reads generated are from the ion torrent (amplicons) method. After performing de novo assembly using Trinity tool, produced contigs are in the form of a single fasta file. The next step is to construct a draft genome from the contigs. Is there any tools/software to create a draft genome from the contigs. Kindly recommend your advice and suggestions and suitable methodology.

Response highly appreciated. Thank you

Example 7 — Bulk Bacterial Genome Assembly

Biostars #417508 · January 20, 2020 · biostars.org/p/417508

Hi,

I am a freshman in sequencing data analysis. When I have one fastq file for only one bacteria, I know how to assemble using SPAdes. For example,

```
spades.py --pe1-1 name.fq.gz --pe1-2 name.fq.gz -o spades_test .
```

But I don't know how to deal with a large number of samples with one Linux command. For example, when I have 10 fastq data (name1~name10), I won't like to assemble them one by one by hand.

Can you tell me how can I do? Thanks!

Example 8 — Taxonomy Classification (MEGAN vs. Kraken)

Biostars #130306 · February 10, 2015 · biostars.org/p/130306

Hi,

I am trying to do taxonomy classification for my unmapped (to reference genome) reads. I tried both Megan and Kraken.

In Kraken I can use read files directly as the input, whereas in Megan I have to make the blastN file first from the read files to give it as the input. No plotting tool is readily available for Kraken, while Megan already has a plotting tool.

I used bacteria, virus and fungi genomes as reference database. I am trying to figure out if I can use read files directly as an input to MEGAN so that I can skip the BLASTN step. I also want to compare the Megan and Kraken output.

Thanks, Deepthi

Example 9 — Average Depth per Amplicon

Biostars #309673 · April 17, 2018 · biostars.org/p/309673

Hello,

I am working with targeted amplicon sequencing data for 200 samples! I want to calculate the average depth per amplicon? I tried bedtools coverage, but it gives total number of reads which map to the target region and I am not sure if that is the average depth. It is little confusing to me.

command used:

```
bedtools coverage -a amplicon_coords.bed -b sample1_RG_sorted_indexed.bam
```

Is there any other way or any modified way to use bedtools coverage option to calculate the average depth per amplicon? Any help will be appreciated. Thanks!!

Example 10 — Comparative Methylation, Two *E. coli* Strains

Biostars #9481497 · July 21, 2021 · biostars.org/p/9481497

Hello,

I am looking to identify differences in methylation between two isogenic replicates of *E. coli*. I have Oxford Nanopore sequencing of three biological replicates for two strains, and am looking to compare methylation between them.

I have done nanopolish analysis and visualized the methylation frequency using ggplot2 in R, and have found no differences in the CpG methylation.

Are there any tools I could use to definitively compare methylation between two strains? Ideally this would generate a PCA plot showing distinct/overlapping methylation profiles. Thank you for any advice!

What ten real examples show

Every one of these: a **concrete data description** (organism, platform, read/sample count) and a **stated goal** — even though none of these writers were told to phrase it that way, across a 7-year span (2014–2021) and a wide range of sub-fields (assembly, typing, taxonomy, amplicon QC, epigenetics, viral variants).

That's the shape 54.6% of real bacteria/virus questions take, and it's exactly the shape `route.py` and the rest of this pipeline are built to consume — not a hypothetical input format, a documented empirical pattern.